AMANDA COSTON

(current as of May 2026)

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Employment

2024-	UNIVERSITY OF CALIFORNIA, BERKELEY *Assistant Professor*, Department of Statistics Faculty, Berkeley Artificial Intelligence Research (BAIR)

Education

2017-2023	CARNEGIE MELLON UNIVERSITY Ph.D. in Machine Learning and Public Policy Advisors: Alexandra Chouldechova & Edward Kennedy Thesis: "Principled machine learning for societally consequential decision making". Committee: Edward Kennedy, Alexandra Chouldechova, Hoda Heidari, & Sendhil Mullainathan
2017-2019	CARNEGIE MELLON UNIVERSITY M.S. in Machine Learning.
2009-2013	PRINCETON UNIVERSITY B.S.E. *magna cum laude* in Computer Science Certificate in the Princeton School of Public and International Affairs Advisor: Robert Schapire Thesis: "Machine learning techniques for the diagnosis of pediatric tuberculosis".

Selected Awards & Honors

Research

2025	Okawa Foundation Research Grant Recipient
2024	Schmidt Sciences AI2050 Early Career Fellow
2023	CMU School of Computer Science Dissertation Award Honorable Mention
2023	Best Paper Award at ACM Conference on Fairness, Accountability, and Transparency (FAccT)
2023	William W. Cooper Doctoral Dissertation Award
2023	Best Paper Award at IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)
2022	Rising Star in EECS by UT-Austin
2022	Rising Star in Machine Learning by University of Maryland
2022	Rising Star in Data Science by University of Chicago
2022	Meta Research PhD Fellow
2022	Future Leader in Responsible Data Science by University of Michigan Institute for Data Science
2020	K&L Gates Presidential Fellow in Ethics and Computational Technologies
2019	NSF Graduate Research Fellow
2019	Tata Consultancy Services Presidential Fellow
2019	Suresh Konda Best First Paper Award by Heinz College of Carnegie Mellon University

Service

2020	Carolyn Comer Graduate Student Involvement Award by Carnegie Mellon University
2013	Department of Computer Science Service Award by Princeton University

Research & Industry Experience

2023-2024	MICROSOFT RESEARCH (MSR) NEW ENGLAND *Postdoc researcher*, Machine Learning and Statistics
2021	FACEBOOK AI APPLIED RESEARCH (FAIAR) *Research intern*, Responsible AI
2020	REGLAB, STANFORD UNIVERSITY *Research Fellow*, Regulation, Evaluation, and Governance Lab at Stanford Law School
2018	IBM RESEARCH AI *Science for Social Good Fellow*
2017	HIVISASA *Technical Consultant*, Kenya
2015-2017	TENEO *Data Scientist*
2013-2015	MICROSOFT *Program Manager*, Bing
2010-2011	SHELTON PSYCHOLOGY LAB, PRINCETON UNIVERSITY *Research Assistant*

Research Interests

Theory:	causal inference, machine learning, algorithmic fairness & societal impacts
Application:	child welfare, consumer credit lending, criminal justice, health policy

Publications & Manuscripts

** indicates joint lead authors*

Working Papers

Garcia H*, Nwankwo E*, Coston A. Sociotechnical Problems in Suicide Prevention in Child Welfare. *(In preparation)*. 2026.

Nwankwo E and Coston A. Seeing Through Noise: Counterfactual Evaluation Under Outcome Measurement Error. *(In preparation)*. 2026.

Liu LT, Raji ID, Zhou A, Guerdan L, Hullman J, Malinsky D, Wilder B, Zhang S, Adam H, Coston A, Laufer B, et al. Bridging Prediction and Intervention Problems in Social Systems. arxiv.org:2507.05216

Ulichney, A and Coston A. Beyond the Training Distribution: Evaluating Predictions under Distribution Shift and Selection Bias. *(Under review)*. 2026.

Coston A, Kennedy EH. Counterfactual audit of racial bias in police traffic stops. *American Causal Inference Conference (ACIC) 2022* oral presentation (20% selection rate).

Coston A, Kennedy EH. The role of the geometric mean in case-control studies. arxiv.org:2207.09016

Publications

Coston A. Falsifying the Discriminant Validity of Predictive Algorithms. *To Appear in Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT).* 2026; doi:10.1145/3805689.3812310. arxiv.org:2601.17146

Morimoto J. and Coston A. Judging the Judges: Evaluating Pretrial Release Decisions with a Simple Nonparametric Approach. *To Appear in Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2026;

Rambachan A, Coston A, Kennedy EH. Robust design and evaluation of predictive algorithms under unmeasured confounding. *The Review of Economics and Statistics (To appear)* arxiv.org:2212.09844

Guerdan L, Coston A, Wu ZS, Holstein K. Predictive performance comparison of decision policies under confounding. *Proceedings of the International Conference on Machine Learning (ICML).* 2024; 16673-16705. http://proceedings.mlr.press/... (arxiv.org:2404.00848)

Kawakami A, Coston A, Zhu H, Heidari H, Holstein K. The Situate AI Guidebook: Co-designing a toolkit to support multi-stakeholder early-stage deliberations around public sector AI proposals. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 2024. doi:10.1145/3613904.3642849. (arxiv.org:2402.18774)

Kawakami A, Coston A, Heidari H, Holstein K, Zhu H. Studying up public sector AI: Shifting our gaze upwards to study systems of power in public sector AI. *SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)*. 2024. doi:10.1145/3686989. (arxiv.org:2405.12458)

Guerdan L, Coston A, Wu ZS, Holstein K. Counterfactual decision support under outcome measurement error. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2023; 1584–1598. doi:10.1145/3593013.3594101. (arxiv.org:2302.11121) **Best Paper Award** by FAccT

Field A, Coston A, Gandhi N, Chouldechova A, Putnam-Hornstein E, Steier D, Tsvetkov Y. Examining risks of racial biases in NLP tools for Child Protective Services. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2023; 1479–1492. doi:10.1145/3593013.3594094

Guerdan L, Coston A, Wu ZS, Holstein K. Ground(Less) truth: A causal framework for proxy labels in human-algorithm decision making. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2023; 688–704. doi:10.1145/3593013.3594036 (arxiv.org:2302.06503)

Coston A, Kawakami A, Zhu H, Holstein K, Heidari H. A validity perspective on evaluating the justified use of data-driven decision-making algorithms. *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. 2023. (arxiv.org:2206.14983). **Best Paper Award** by SaTML

Coston A*, Rambachan A*, Chouldechova A. Characterizing fairness over the set of good models under selective labels. *Proceedings of the International Conference on Machine Learning (ICML)*. 2021; 2144-2155. http://proceedings.mlr.press/... (arxiv.org:2101.00352)

Coston A, Guha N, Ouyang D, Lu L, Chouldechova A, Ho DE. Leveraging administrative data for bias audits: Assessing disparate coverage with mobility data for COVID-19 policy. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2021; 173-184. doi:10.1145/3442188.3445881 (arxiv.org:2011.07194)

Coston A, Kennedy EH, Chouldechova A. Counterfactual predictions under runtime confounding. *Advances in Neural Information Processing Systems 33 (NeurIPS)*. 2020; 4150-4162. https://papers.nips.cc/paper/... (arxiv.org:2006.16916)

Coston A, Mishler A, Kennedy EH, Chouldechova A. Counterfactual risk assessments, evaluation, and fairness. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2020; 582-593. doi:10.1145/3351095.3372851 (arxiv.org:1909.00066)

Zhao H, Coston A, Adel T, Gordon GJ. Conditional learning of fair representations. *International Conference on Learning Representations (ICLR)*. 2020. https://iclr.cc/... (arxiv.org:1910.07162)

Li L, Zuo R, Coston A, Weiss JC, Chen GH. Neural topic models with survival supervision: Jointly predicting time-to-event outcomes and learning how clinical features relate. *International Conference on Artificial Intelligence in Medicine (AIME)*. 2020; 371-381. https://link.springer.com/... (arxiv.org:2007.07796)

Coston A, Ramamurthy KN, Wei D, Varshney KR, Speakman S, Mustahsan Z, Chakraborty S. Fair transfer learning with missing protected attributes. *Proceedings of the AAAI / ACM Conference on Artificial Intelligence, Ethics, and Society (AIES)*. 2019; 91-98. doi:10.1145/3306618.3314236

Book Chapter

Coston A, Rubio MD, Kennedy EH. Statistical analysis of randomized experiments. *AI for Social Impact*. ai4sibook.org

Peer-reviewed non-archival papers

Ulichney, A and Coston A. Double Machine Learning Evaluation under Distribution Shift and Selection Bias. *NeurIPS 2025 Workshop on Reliable ML from Unreliable Data.* https://neurips.cc/.../125256

Kawakami A, Coston A, Heidari H, Holstein K, Zhu H. Studying up public sector AI: Shifting our gaze upwards to study systems of power in public sector AI. *ACM conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO 2023).* oral presentation (18% selection rate)

Kawakami A, Coston A, Zhu H, Heidari H, Holstein K. Recentering validity considerations through early-stage deliberations around AI and policy design. *CHI 2023 Workshop on Designing Technology and Policy*.

Rambachan A, Coston A, Kennedy EH. Counterfactual risk assessments under unmeasured confounding. *ACIC 2022*. *NeurIPS 2022 Workshop on Algorithmic Fairness through the Lens of Causality and Privacy.* arxiv.org:2212.09844

Guerdan L, Coston A, Wu ZS, Holstein K. Counterfactual decision support under treatment-conditional outcome measurement error. *NeurIPS 2022 Workshop on Causality for Real-world Impact.*

Guerdan L, Coston A, Wu ZS, Holstein K. Ground(less) truth: The problem with proxy outcomes in human-AI decision making. *NeurIPS 2022 Workshop on Human-Centered AI.*

Coston A, Kennedy EH. Counterfactual audit of racial bias in police traffic stops. *ACIC 2022* oral presentation (20% selection rate).

Coston A, Kawakami A, Zhu H, Holstein K, Heidari H. A validity perspective on evaluating the justified use of data-driven decision-making algorithms. *ACM conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO 2022)*. arxiv.org:2206.14983

Coston A, Kennedy EH, Chouldechova A. Counterfactual risk assessments, evaluation, and fairness. *NeurIPS 2019 Workshop on Causal Machine Learning.*

Coston A, Kennedy EH, Chouldechova A. Counterfactual risk assessments and evaluation for child welfare screening. *ACIC 2019*.

Coston A, Leqi L. Offline heterogeneous policy evaluation: A causal approach. *ICML 2018 Workshop on Causal ML.*

Presentations

** indicates presentation scheduled for future date*

Invited Talks

2026	Statistical Methods for Health Equity webinar, University College London
2026	Machine Learning in Econometrics (Econ 241A) Guest Lecture, University of California, Berkeley
2026	Statistics Seminar, University of California, Santa Cruz
2026	Bridging Prediction and Intervention Problems in Social Systems, Simons Institute for the Theory of Computing, Berkeley, CA
2024	Berkeley Statistics Annual Research Symposium, Berkeley, CA
2024	Stanford Statistics Seminar, Palo Alto, CA
2024	Methods Workshop, UC Berkeley, Berkeley, CA
2024	Biostatistics Seminar, UC Berkeley, Berkeley, CA
2024	Statistical Approaches to Fair Decision Making, Joint Statistical Meetings, Virtual
2024	Schmidt Sciences AI2050 Fellows Summit, Palo Alto, CA

2024 Bridging Prediction and Intervention Problems in Social Systems, Banff International Research Station (BIRS), Banff, Canada
2024 Toronto Data Workshop, Virtual
2024 Workshop on Operationalizing NIST AI RMF, Northeastern University, Boston, MA
2024 Applied Statistics Workshop, Department of Government and Institute for Quantitative Social Science, Harvard University, Boston, MA
2024 Econometrics Seminar, Department of Economics, Harvard University, Boston, MA
2023 Bringing Statistical Thinking into Fair, Transferable Machine Learning, Institute of Mathematical Sciences International Conference on Statistics and Data Science, Lisbon, Portugal
2023 Department of Human Services: Analytics, Technology, and Planning, Allegheny County, Pittsburgh, PA
2023 Theory of Computing for Fairness (TOC4Fairness) Seminar Series, Simons Foundation, Virtual
2023 Public Policy and the AI Revolution, Association for Public Policy Analysis & Management, Atlanta, GA
2023 Quantitative Methods Workshop, Yale University, New Haven, CT
2023 Statistically Significant: Equity Concerns in Algorithmic Bias, Privacy, and Survey Representation, Joint Statistical Meetings, Toronto, Canada
2023 K&L Gates Conference in Ethics and AI, Carnegie Mellon University, Pittsburgh, PA
2023 Multigroup Fairness and the Validity of Statistical Judgment, Simons Institute for the Theory of Computing, Berkeley, CA
2023 Automated Decision Systems Reading Group, University of California, Berkeley, CA
2023 Center for Information Technology Policy Lecture, Princeton University, Princeton, NJ
2023 Department of Computer Science, George Mason University, Fairfax, VA
2023 AI Seminar, New York University, New York, NY
2023 Data Science Initiative seminar, Brown University, Providence, RI
2023 Department of Engineering and Public Policy seminar, Carnegie Mellon University, Pittsburgh, PA
2023 Khoury College of Computer Sciences Lecture, Northeastern University, Boston, MA
2023 Department of Computer Science, University of Maryland, College Park, MD
2023 Halicioglu Data Science Institute and the School of Global Policy and Strategy, University of California, San Diego, CA
2023 Department of Statistics, University of California, Berkeley, CA
2023 McCourt School of Public Policy, Georgetown University, Washington, D.C.
2023 Information Science Colloquium Series, Cornell University, Ithaca, NY
2023 The Division of Decision, Risk, and Operations, Columbia Graduate School of Business, New York, NY
2023 The School of Data Science Colloquium, University of Virginia, Charlottesville, VA
2023 Econometrics & Statistics group, University of Chicago Booth School of Business, Chicago, IL
2022 Operations, Information, and Decisions Department, Wharton School of the University of Pennsylvania, Philadelphia, PA
2022 Symposium on Frontiers of Machine Learning & AI, University of Southern California, LA, CA
2022 INFORMS Session on Finding Sets of Near-Optimal Solutions for Mixed-Integer Programs, Indianapolis, IN
2022 Brown University Bravo Center Workshop on the Economics of Algorithms, Providence, RI
2022 Stanford University RegLab Summer Institute Speaker Series, Virtual
2021 Merck Data Science All Hands, Virtual
2021 Johns Hopkins University Causal Inference Working Group, Virtual
2021 PlaceKey COVID-19 Data Consortium, Virtual
2021 University of Pennsylvania Department of Biostatistics and Epidemiology, Virtual
2020 University of Chicago Crime Lab, Virtual

Doctoral Consortia

2022	EAAMO (ACM conference on Equity & Access in Algorithms, Mechanisms, and Optimization)
2022	FAccT (ACM Conference on Fairness, Accountability, and Transparency)
2020	FAccT (ACM Conference on Fairness, Accountability, and Transparency)
2019	AIES (AAAI / ACM Conference on Artificial Intelligence, Ethics, and Society)

Patents

2022	Enhancing Fairness in Transfer Learning for Machine Learning Models with Missing Protected Attributes in Source or Target Domains. Supriyo Chakraborty, Amanda Coston, Zairah Mustahsan, Karthikeyan Natesan Ramamurthy, Skyler Speakman, Kush R. Varshney, and Dennis Wei. US 11,443,236. *Granted.*

Service

Organization

2019-2023	Steering Committee of Machine Learning for Developing World (ML4D) NeurIPS Workshop
2019-2020	Co-organizer of Fairness, Ethics, Accountability, and Transparency Reading Group at CMU
2019-2020	Co-organizer of Machine Learning Department (MLD) Tea at CMU
2018-2019	Co-organizer of ML4D NeurIPS Workshop

Journal Referee

Nature Human Behaviour
Journal of the Royal Statistical Society (JRSS-B)
Journal of the American Statistical Association (JASA)
Transactions on Machine Learning Research
Data Mining and Knowledge Discovery
Harvard Data Science Review

Program Committee and Conference Reviewer

2026	Reviewer, NeurIPS
2026	Program Committee, FAccT
2024	Reviewer, ICML
2023	Ethical Reviewer, NeurIPS
2023	Program Committee, EAAMO
2023	Reviewer, ICLR
2022	Ethical Reviewer, NeurIPS
2022	Reviewer, NeurIPS
2022	Reviewer, NeurIPS Datasets and Benchmarks
2022	Program Committee, EAAMO
2022	Program Committee, FAccT
2022	Reviewer, ICML
2022	Reviewer, ICLR
2021	Area Chair, Responsible AI workshop at ICLR
2021	Ethical Reviewer, NeurIPS
2021	Reviewer, NeurIPS

2021	Reviewer, NeurIPS Datasets and Benchmarks
2021	Program Committee, FAccT
2021	Reviewer, ICML
2020	Reviewer, NeurIPS
2020	Program Committee, FAccT
2020	Reviewer, ICML
2020	Program Committee, AIES
2020	Program Committee, AAAI Emerging Track on AI for Social Impact
2019	Program Committee, IJCAI Workshop on AI for Social Good

Leadership

2012-2013	Committee on Discipline, Princeton University
2012-2013	Computer Science Undergraduate Council, Princeton University

Invited Conference & Workshop Roles

2022	Roundtable Lead for NeurIPS Workshop on Algorithmic Fairness through Lens of Causality
2022	Breakout Group Moderator for CCC & INFORMS Workshop II on AI/OR
2022	Breakout Group Moderator for NSF-Amazon Fairness in AI Principal Investigator meeting
2022	Session Chair for Responsible Data Management Session at FAccT

Teaching Experience

Instructor

2026 Spring	Statistical Models: Theory and Application (STAT 215b), UNIVERSITY OF CALIFORNIA, BERKELEY
2024 Fall	Causal Inference (STAT 156/256), UNIVERSITY OF CALIFORNIA, BERKELEY

Teaching Assistant

2021 Spring	Introduction to Machine Learning (10-301/10-601), CARNEGIE MELLON UNIVERSITY
2012 Fall	Computers in our World (COS 109), PRINCETON UNIVERSITY

Project Instructor

2019 Summer	AI4ALL, CARNEGIE MELLON UNIVERSITY ▷ Developed and led a project on algorithms, criminal justice, & fairness for high schoolers from historically excluded communities.

Mentorship

2022-2023	Women@SCS Mentor
2019-2023	CMU AI Mentor
2019	Women@SCS Roundtable Leader

2016-2017	Read Ahead Mentor
2014-2015	MySkills4Afrika (Microsoft) Virtual Mentor

Hackathon Distinctions

2015	Microsoft OneWeek Hackathon, Bing Finalist ▷ Web answer to enable victims of revenge porn to remove content from Bing and OneDrive
2013	NYU-Abu Dhabi Hackathon for the Social Good, 2nd Place ▷ Android app for sharing a travel route to facilitate safe travel for women
2012	Tiger Launch, Social Entrepreneurship, 3rd Place ▷ Web service using QR codes to empower consumers to support value-aligned businesses

Civic Engagement

2014-2015	Court Appointed Special Advocate, Family Law CASA ▷ Represented the child's interest in family law cases
2010-2012	Engineers Without Borders ▷ Obtained & configured 50 One Laptop Per Child netbooks for a library in Ashaiman, Ghana
2007-2008	Congressional Intern, U.S. House of Representatives ▷ Office of Congressman John Spratt representing South Carolina's 5th congressional district

Media Coverage

2021	”Smartphone Location Data Can Leave Out Those Most Hit by Covid-19.” *Wall Street Journal*. https://www.wsj.com/articles/...
2020	”Stanford and Carnegie Mellon find race and age bias in mobility data that drives COVID-19 policy.” *VentureBeat*. https://venturebeat.com/ai/...